



**BAULKHAM HILLS
HIGH SCHOOL**

HSC TASK 1 2020

Mathematics Advanced

**General
Instructions**

- Reading time – 5 minutes
- Working time – 1 hour
- Write using black pen
- NESA approved calculators may be used
- In Questions 6 – 14, show relevant mathematical reasoning and/or calculations

**Total marks:
37**

Section I – 5 marks (pages 2–3)

- Attempt Questions 1 – 5
- Allow about 8 minutes for this section

Section II – 32 marks (Answer Booklet pages 5–15)

- Attempt Questions 6 – 14
- Allow about 52 minutes for this section

Section I

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 5.

1 The limiting sum of an infinite geometric series exists when:

- A. $r > 1$
- B. $|r| > 1$
- C. $-1 < r < 1$
- D. $r < 1$

2 What is the domain of the function $f(x) = \sqrt{1 - x^2}$

- A. $(0, 1)$
- B. $[0, 1]$
- C. $(-1, 1)$
- D. $[-1, 1]$

- 3 Which of the following pairs of transformations commute?
- A. Vertical reflection and vertical translation.
 - B. Horizontal translation and horizontal dialation.
 - C. Reflection in the y –axis and reflection in the x –axis.
 - D. Vertical translation and vertical dialation.
- 4 The function $y = 3\cos 2x - 5$ has :
- A. Amplitude 3, period 2 and mean value -5
 - B. Amplitude 5, period $\frac{1}{2}$ and mean value 3
 - C. Amplitude 3, period π and mean value 5
 - D. Amplitude 3, period π and mean value -5
- 5 Consider the series $\sqrt{5} + \sqrt{45} + \sqrt{125} + \dots + z = 225\sqrt{5}$, the value of z is:
- A. $25\sqrt{5}$
 - B. $29\sqrt{5}$
 - C. $30\sqrt{5}$
 - D. $35\sqrt{5}$

End of Section I

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2020**HSC TASK 1**

Mathematics Advanced

Section II Answer Booklet

32 marks**Attempt Questions 6 – 14****Allow about 52 minutes for this section**

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Question 6 (5 marks)

A gardener plants a bed of roses. The bed is planted so that the first row has 24 rose plants. The second row has 29 rose plants. Each succeeding row has 5 more rose plants than the previous row.

a) Calculate the number of rose plants in the 8th row.

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b) Which row would be the first to contain more than 150 rose plants?

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c) The gardener has planted 2895 rose plants altogether. Assuming that the above pattern has been continued, how many rows were planted?

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Question 7 (5 marks)

- (a) Solve the equation $x^2 - 6 = x$

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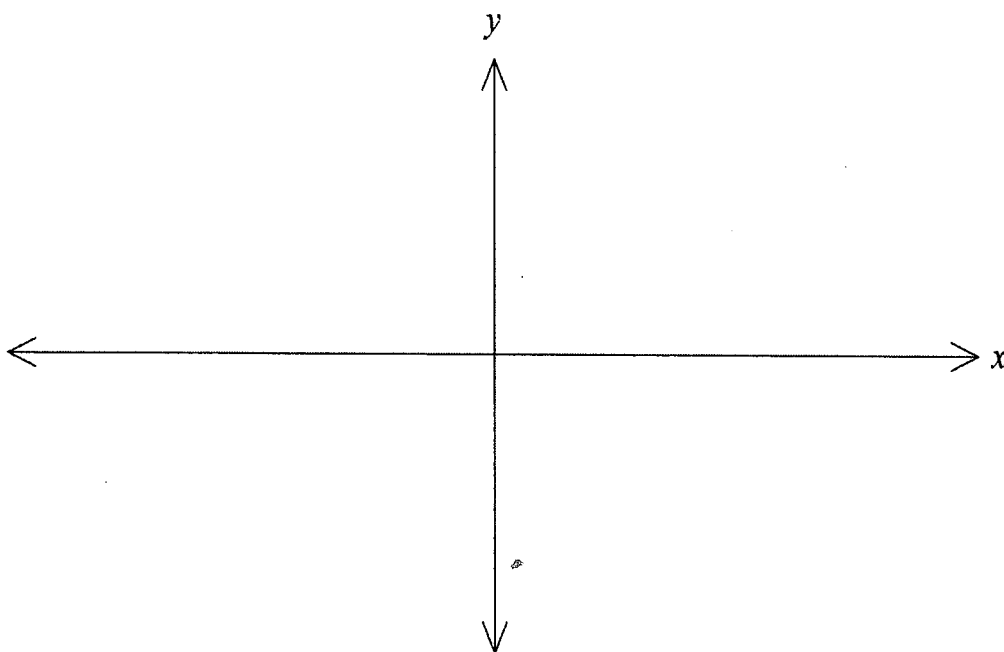
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- (b) In the space below, sketch the graph of $y = x^2 - 6$ and $y = x$. Label the coordinates of any point of intersection and any x or y intercepts of the graph.

2



- (c) Find the values of x , for which $x^2 - 6 > x$.

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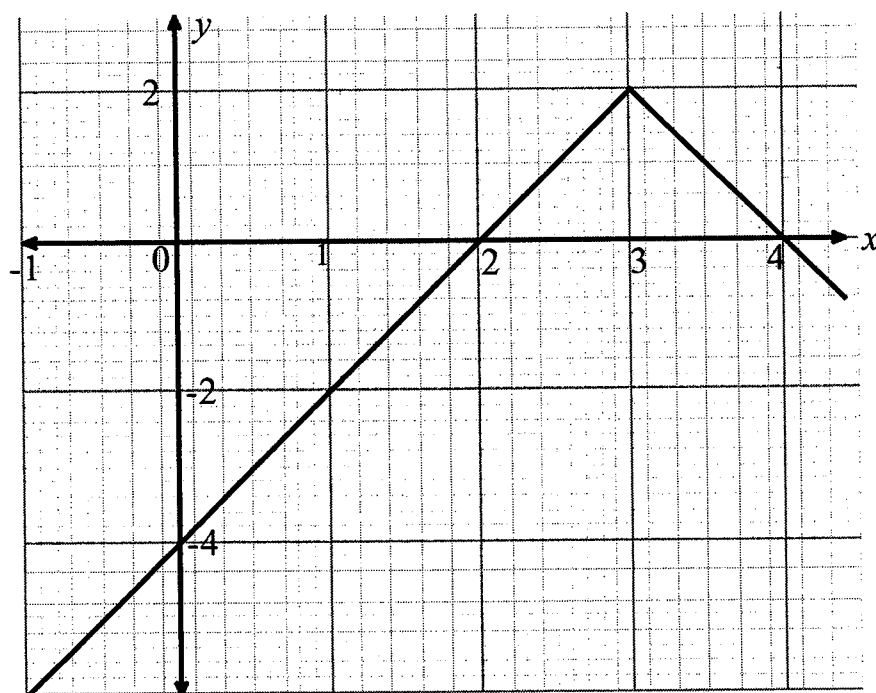
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Question 8 (3 marks)

The function $f(x) = |x|$ is transformed and the equation of the new function is of the form $y = kf(x + b) + c$, where k , b and c are constants. The graph of the new function is shown below. 3



What are the values of k , b and c ?

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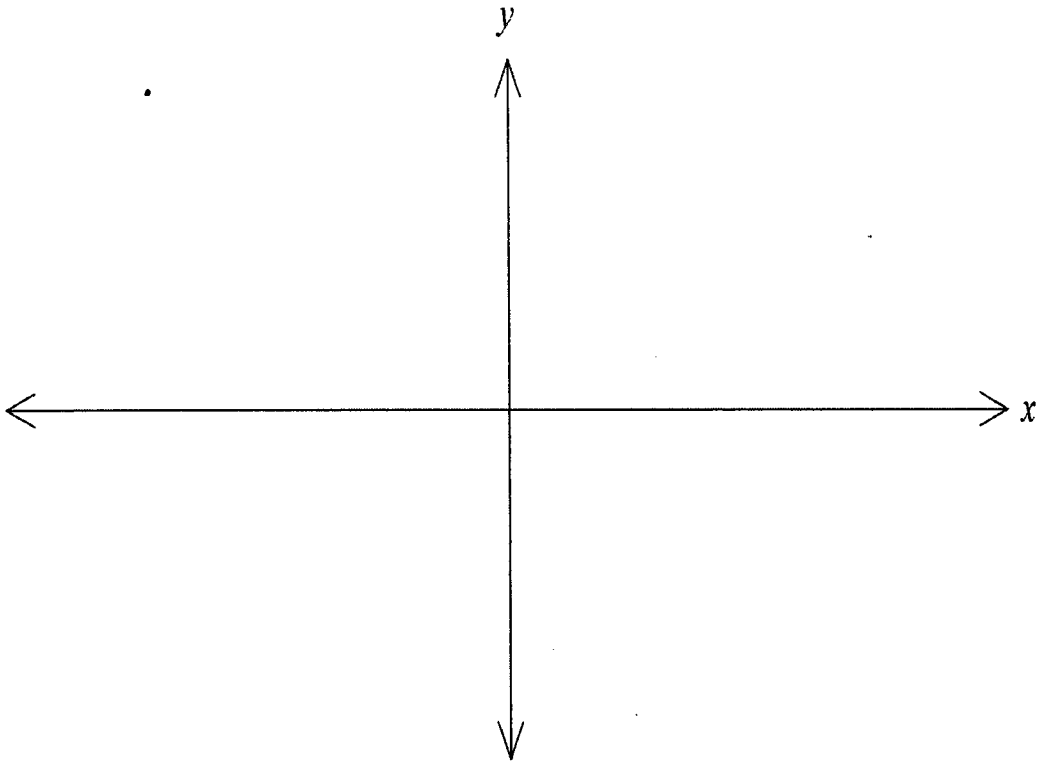
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Question 9 (3 marks)

- (a) Sketch $y = 3\cos x - 2$ and $y = \frac{x}{4} - 1$ in the domain $0 \leq x \leq 2\pi$. 2



- (b) Hence or otherwise, how many solutions are there to the equation 1

$$3\cos x - 2 = \frac{x}{4} - 1?$$

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Question 11 (5 marks)

Consider the function $y = \frac{8}{x^2-4}$

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- (a) Write down the domain.

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- (b) Find the vertical and horizontal asymptotes

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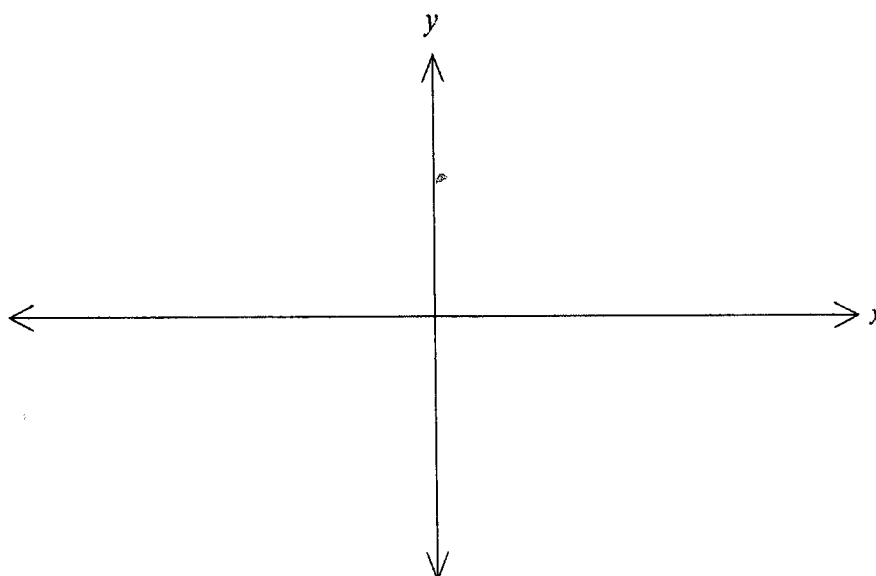
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- (c) Hence sketch the graph of $y = \frac{8}{x^2-4}$, showing all intercepts and asymptotes.

2

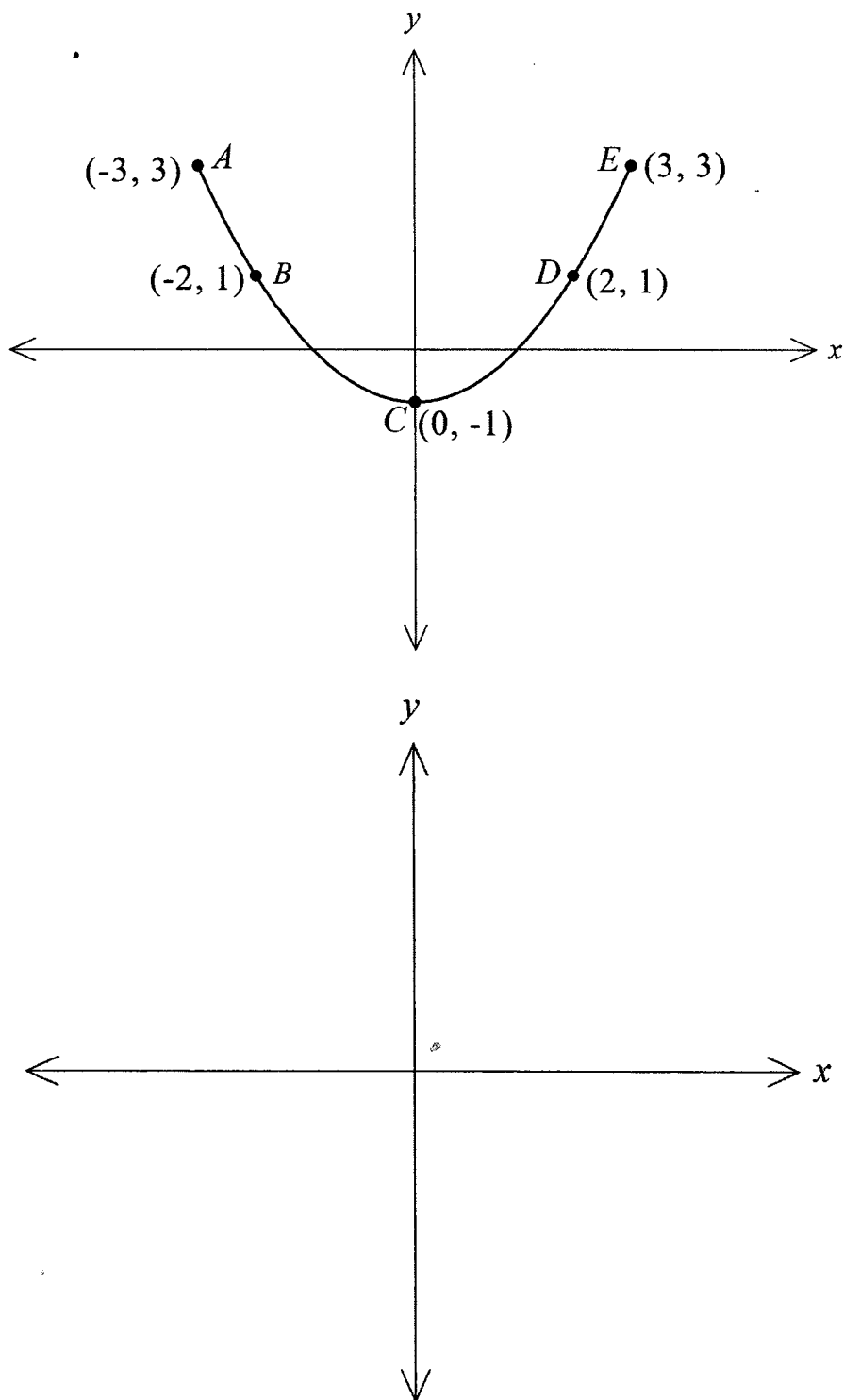


In a geometric progression $S_4 = 30$, while the sum to infinity is 32. Find the first three terms.

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Question 13 (3 marks)

Given the sketch of the function $y = f(x)$, sketch the graph of $y = 3f(x + 3)$, clearly indicating the new coordinates for A, B, C, D and E on the graph. 3



Question 14 (3 marks)

For a given arithmetic sequence, the ratio of the sum of the first m terms to the sum of first n terms is $m^2 : n^2$, where $m \neq n$.

Prove that the ratio of the m^{th} term to the n^{th} term is $(2m - 1):(2n - 1)$.

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END OF EXAM



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- A. $25\sqrt{5}$
- ☒ B. $29\sqrt{5}$
- C. $30\sqrt{5}$
- D. $35\sqrt{5}$

$$\begin{aligned} \sqrt{5} + 3\sqrt{5} + 5\sqrt{5} + \dots + z &= 225\sqrt{5} \\ a &= \sqrt{5}, d = 2 \\ 1 + 3 + 5 + \dots + n &= 225 \rightarrow a=1, d=2 \\ n &= a + (n-1)d \\ &= 1 + (n-1) \times 2 \\ &= 2n-1 \\ S_n &= 225 = \frac{n}{2} [a + l] \\ &= \frac{n}{2} [1 + 2n-1] \\ 225 &= n^2 \\ n &= \pm 15 \quad n > 0 \\ \therefore n &= 15 \\ 2\sqrt{5} (2n-1) \sqrt{5} &= (2 \times 15-1) \sqrt{5} \\ &= 29\sqrt{5} \end{aligned}$$

End of Section I

Question 6 (5 marks)

A gardener plants a bed of roses. The bed is planted so that the first row has 24 rose plants. The second row has 29 rose plants. Each succeeding row has 5 more rose plants than the previous row.

a) Calculate the number of rose plants in the 8th row.

$$24, 29, 34 \dots$$

1-correct answer

$$\begin{aligned} T_8 &= a + (n-1)d \\ &= 24 + (8-1) \times 5 \\ &= 59 \end{aligned}$$

b) Which row would be the first to contain more than 150 rose plants?

2

$$T_n > 150$$

2- correct solution

$$24 + (n-1)d > 150$$

1- Establishing $T_n > 150$

$$n-1 > \frac{126}{5}$$

$$n > 25.2 + 1 = 26.2$$

$$n = 27$$

c) The gardener has planted 2895 rose plants altogether. Assuming that the above pattern has been continued, how many rows were planted?

2

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

2-correct solution

$$2895 = \frac{n}{2} (48 + 5n - 5)$$

1-forming the quadratic equation or

Using the sum formula

$$5790 = \frac{n}{2} (43 + 5n)$$

$$5n^2 + 43n - 5790 = 0$$

$$n = \frac{-43 \pm \sqrt{43^2 - 4 \times 5 \times -5790}}{10}$$

$$= \frac{-43 \pm 343}{10}$$

$$n = 30 \text{ rows as } n > 0$$

Question 7 (5 marks)

- (a) Solve the equation $x^2 - 6 = x$

$$x^2 - x - 6 = 0$$

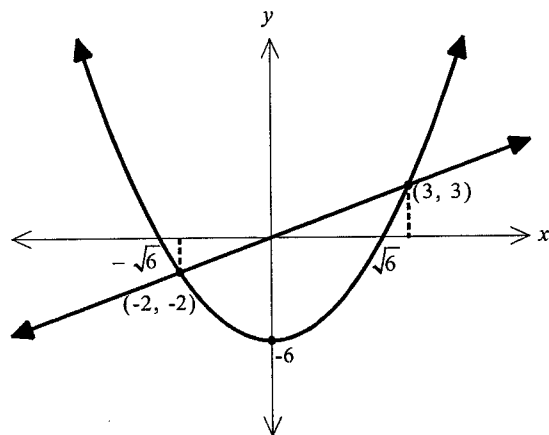
$$(x - 3)(x + 2) = 0$$

$$x = 3 \text{ or } x = -2$$

1- correct solution

1

- (b) In the space below, sketch the graph of $y = x^2 - 6$ and $y = x$. Label the coordinates of any point of intersection and any x or y intercepts of the graph.



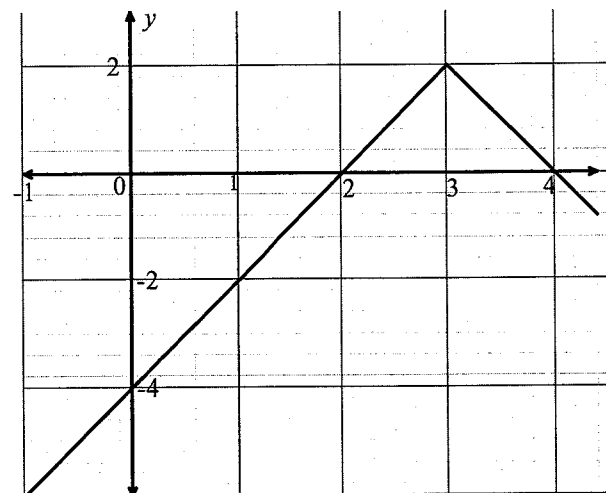
2-Correct graphs with point of intersection

1- Correct graph without the label

2

Question 8 (3 marks)

The function $f(x) = |x|$ is transformed and the equation of the new function is of the form $y = kf(x + b) + c$, where k, b and c are constants. The graph of the new function is shown below.



What are the values of k, b and c ?

$f(x) = |x|$ is translated to the right by 3 units. Reflected through the x -axis, multiplied by 2 and translated up by 2 units.

3- all correct answer

2- two correct answers

1- One correct answer

$$\therefore k = -2$$

$$b = -3$$

$$c = 2$$

- (c) Find the values for which $x^2 - 6 > x$.

$$x < -2 \text{ or } x > 3$$

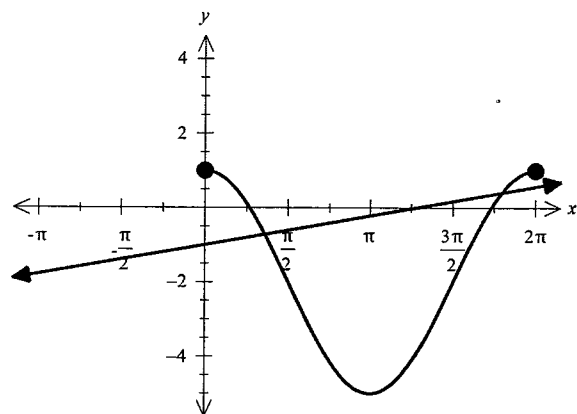
2-correct solutions

1-one correct inequality.

2

Question 9 (3 marks)

- (a) Sketch $y = 3\cos x - 2$ and $y = \frac{x}{4} - 1$ in the domain $0 \leq x \leq 2\pi$.



2- Both the graphs are correct.
1- One of the them is correct

2

Question 10 (2 marks)

Express $0.4\dot{6}$ as infinite series and hence express as simplest fraction.

2

$$\begin{aligned} &0.4666 \dots \\ &= 0.4 + 0.06 + 0.006 + \dots \\ &= 0.4 + \frac{6}{100} + \frac{6}{1000} + \dots \\ &= 0.4 + \frac{\frac{6}{100}}{1 - \frac{1}{10}} \\ &= \frac{4}{10} + \frac{\frac{6}{100}}{\frac{9}{10}} \\ &= \frac{4}{10} + \frac{2}{30} \\ &= \frac{14}{30} \\ &= \frac{7}{15} \end{aligned}$$

2-correct solution

1- Forming the series correctly/
recognising a and r .

- (b) Hence or otherwise, how many solutions are there to the equation

1

$$3\cos x - 2 = \frac{x}{4} - 1?$$

2 solutions

1- Correct answer

Question 11 (5 marks)

Consider the function $y = \frac{8}{x^2-4}$.

(a) Write down the domain.

Domain : all real values of $x, x \neq \pm 2$

1- Correct answer

(b) Vertical asymptotes: $x = \pm 2$

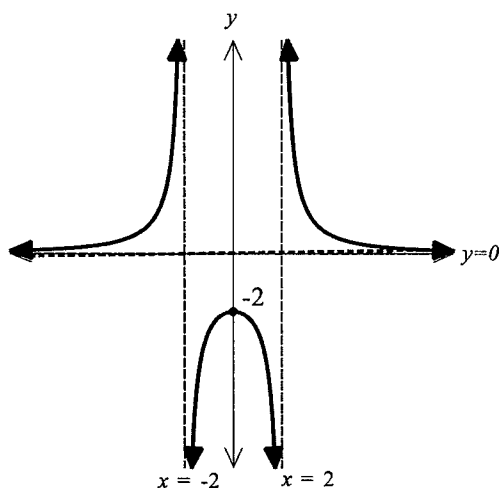
2- Both correct asymptotes

1- One correct asymptote

Horizontal asymptote: $\lim_{x \rightarrow \pm \infty} \frac{8}{(x^2-4)} = 0$

$\therefore y = 0$

(b) Hence sketch the graph of $y = \frac{8}{x^2-4}$, showing all intercepts and asymptotes.



2- Correct graph with labels

1- Correct asymptotes

Question 12 (3 marks)

$$30 = S_4 = \frac{a(1-r^4)}{1-r}$$

$$32 = S_{\infty} = \frac{a}{1-r}$$

$$a = 32(1-r)$$

$$30 = \frac{32(1-r)(1-r^4)}{1-r}$$

$$1-r^4 = \frac{30}{32}$$

$$r^4 = 1 - \frac{30}{32}$$

$$r^4 = \frac{1}{16}$$

$$r = \pm \frac{1}{2}$$

3- correct solution

2- finding the two values of r

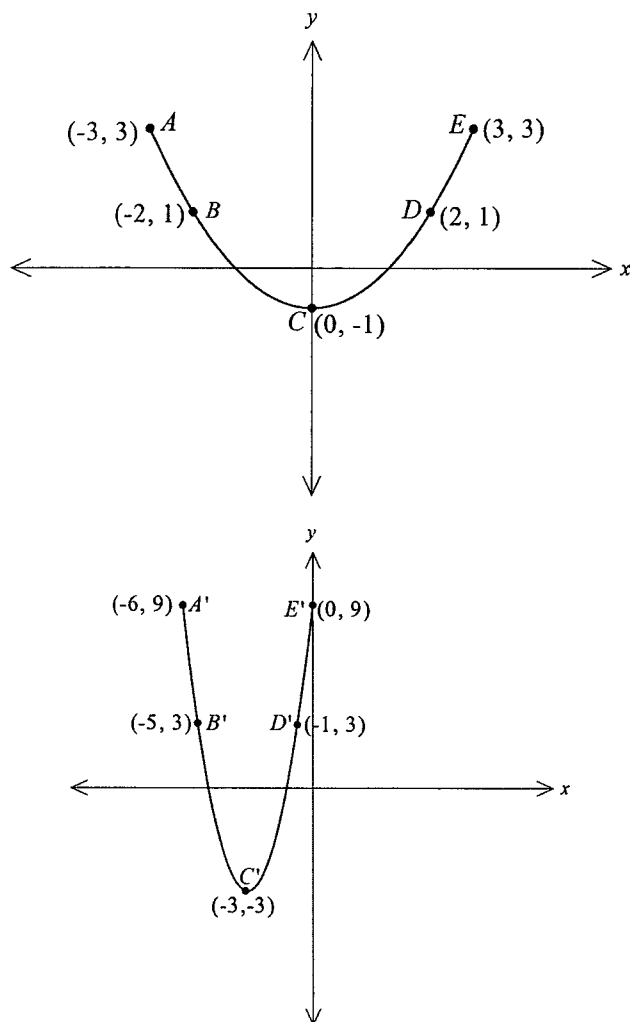
1- finding a in terms of r .

$r = \frac{1}{2}$	$r = -\frac{1}{2}$
$32 = \frac{a}{1-\frac{1}{2}}$	$32 = \frac{a}{1+\frac{1}{2}}$
$= \frac{a}{\frac{1}{2}}$	$32 = \frac{2a}{3}$
$a = 16$	$a = 48$
16, 8, 4...	48, -24, 12.....

Question 13 (3 marks)

Given the sketch of the function $y = f(x)$, sketch the graph of $y = 3f(x + 3)$.

3



- 3- correct graph with label
- 2- Correct horizontal transformation
- with correct vertex and y-intercept
- 1- Correct horizontal transformation

Question 14 (3 marks)

For a given arithmetic sequence, the ratio of the sum of the first m terms to the sum of first n terms is $m^2 : n^2$, where $m \neq n$.

Prove that the ratio of the m^{th} term to the n^{th} term is $(2m - 1) : (2n - 1)$

$$\frac{\frac{m}{2}(2a + (m-1)d)}{\frac{n}{2}(2a + (n-1)d)} = \frac{m^2}{n^2}$$

$$\frac{2a + (m-1)d}{2a + (n-1)d} = \frac{m}{n}$$

- 3- correct solution
- 2-showing $a = 2d$
- 1- sig. progress towards the proof

$$2an + mnd - nd = 2am + mnd - md$$

$$2an - 2am = nd - md$$

$$2a(n - m) = d(n - m)$$

$$d = 2a \text{ ----- (1)}$$

$$\text{From (1)} \quad \frac{a + (m-1)d}{a + (n-1)d} = \frac{a + (m-1)2a}{a + (n-1)2a}$$

$$= \frac{1 + (m-1) \times 2}{1 + (n-1) \times 2}$$

$$= \frac{1 + 2m - 2}{1 + 2n - 2}$$

$$= \frac{2m - 1}{2n - 1}$$

$$(2m - 1) : (2n - 1) \text{ Proved}$$

END OF EXAM